Diagnose Trait Matrix Readiness

Description

Summarises coverage, transformed dynamic range, invalid values, and robust taxon-level extremes in a community trait matrix without changing values.

Usage

```
diagnose_trait_matrix_readiness(
  data_trait_table,
  vec_trait_transformations,
  taxon_column = "taxon_name",
  maximum_missing_fraction = 0.5,
  maximum_transformed_range = 4,
  outlier_fence_multiplier = 3
)
```

Arguments

`data_trait_table`	Wide taxon-by-trait data frame.
`vec_trait_transformations`	Named transformation contract accepted by [prepare_trait_matrix_for_dissimilarity()].
`taxon_column`	Character scalar naming the taxon identifier column.
`maximum_missing_fraction`	Maximum acceptable missing fraction per trait domain.
`maximum_transformed_range`	Maximum acceptable transformed range.
`outlier_fence_multiplier`	Non-negative Tukey-fence multiplier.

Value

A named list containing 'data_domain_summary', 'data_taxon_anomalies', and scalar 'flag_ready'.

Examples

```
diagnose_trait_matrix_readiness(
  data_trait_table = tibble::tibble(
    taxon_name = c("A", "B"),
    `Plant heigh` = c(10, 100)
  ),
  vec_trait_transformations = c(`Plant heigh` = "log10")
)
```